



Vivekananda Institute of Professional Studies – Technical Campus

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SCHOOL OF ENGINEERING AND TECHNOLOGY

SUMMER TRAINING REPORT

Enterprise Information Portal (EIP): A Study at L&T Construction

Submitted in partial fulfillment of the
Requirements for the award of
Degree of Bachelor of Technology
in
Artificial Intelligence and Data Science

Submitted By

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Enrollment No.: 52017711924

Semester: V

Academic Year: 2026–27

DECLARATION

I, Mohit Gupta, bearing Enrollment No. 52017711924, a student of B.Tech. in Artificial Intelligence and Data Science (5th Semester) at Vivekananda Institute of Professional Studies – Technical Campus, hereby declare that the Training Report entitled "Enterprise Information Portal (EIP): A Study at L&T Construction" is an authentic record of my own work carried out during the Summer Internship undertaken at Larsen & Toubro (L&T) Construction, Water & Effluent Treatment, from 15th June, 2026 to 12th August, 2026, as a requirement of the B.Tech. programme, School of Engineering and Technology, VIPS-TC.

To the best of my knowledge, all the data presented in this report is genuine and accurate. This report has not been submitted to any other institution for the award of any degree.

Mohit Gupta
52017711924

Date:

Place:

ACKNOWLEDGEMENT

A training report, from its inception to completion, is shaped by the support and encouragement of many individuals. I take this opportunity to express my heartfelt gratitude to all those who guided and supported me throughout the course of this study.

First and foremost, I would like to extend my sincere thanks to Mr. Harsh Juneja, Head – Career Development Centre, Vivekananda Institute of Professional Studies – Technical Campus, New Delhi, for facilitating my Summer Internship at Larsen & Toubro (L&T) Construction.

I am deeply grateful to my mentor at the organisation, Mr. Anurag Awasthi, for the invaluable support and guidance provided during the internship. His insights and encouragement played a crucial role in the successful completion of this training.

I would also like to express my appreciation to the faculty coordinators and mentors at VIPS-TC for their continuous support, motivation, and assistance in preparing the final report and presentation.

Lastly, my sincere thanks to all the faculty and staff members of the School of Engineering and Technology for their cooperation and for making this training experience enriching and rewarding.

Submitted By:
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INTERNSHIP OFFER LETTER



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June 10, 2026

Mr. Harsh Juneja
Head - Career Development Centre
Vivekananda Institute of Professional Studies Technical Campus

Dear Sir,

This has reference to your request regarding internship training for the following student pursuing B.Tech (AI - DS) from your institution in our organization.

- **Mr. Mohit Gupta**

As a special case, we agree to admit him to do the internship from 15th June, 2026 for a period of 2 months in our organization. He may be advised to report at the following address for internship.

*L&T Construction
Water & Effluent Treatment
Tower II, 11th Floor, 12/4
Delhi Mathura Road, Sector 27D (Adjacent to Sarai Metro Station)
Faridabad, Haryana - 121003
Contact Person: Mr. Anurag Awasthi
Contact no: 9971509222*

Please note that he has to undergo internship training without any financial commitment to the company and fully at his own risk and responsibility. Format of indemnity bond to this effect is enclosed. He may send us the same on stamp paper duly signed by him and notarized.

Thanking you,



Yours faithfully,
for LARSEN & TOUBRO LIMITED

K. Krishna Kumar
(K. KRISHNA KUMAR)
HR DEPARTMENT
WATER & EFFLUENT TREATMENT
CHENNAI

Registered Office: L&T House, N. M. Marg, Ballard Estate, Mumbai - 400 001, INDIA
CIN - L99999MH1946PLC004768

L&T Construction - Water & Effluent Treatment is a brand of Larsen & Toubro Limited

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TO WHOMSOEVER IT MAY CONCERN

Thanking you,

Yours faithfully,
for LARSEN & TOUBRO LIMITED,

(K. KRISHNA KUMAR)

HR DEPARTMENT

WATER & EFFLUENT TREATMENT
CHENNAI



Registered Office: L&T House, N. M. Marg, Ballard Estate, Mumbai - 400 001, INDIA
Sensitivity: LNT Construction Internal Use
Reference No.: CN/L99999MH/1946PLC004768

L&T Construction - Water & Effluent Treatment is a brand of Larsen & Toubro Limited

EMPLOYER / MENTOR FEEDBACK FORM

Name of the Employer: Mr. Anurag Awasthi	Organization: Larsen & Toubro (L&T) Construction
Name of the Intern: Mohit Gupta	Role of the Intern: Summer Trainee, Information System Department

Parameter for Assessment	Highly Satisfied	Satisfied	Moderately Satisfied	Dissatisfied	Highly Dissatisfied
Relevant technical knowledge essential to do his/her assignments.	✓				
Ability to identify, analyze and research literature for assigned problems.	✓				
Ability to exhibit awareness of societal, health, safety, legal and cultural responsibilities relevant to the field.	✓				
Ability to exhibit ethical growth and professionalism.	✓				
Ability to complete assignments, meet deadlines, understand instructions and work with others.	✓				
Oral and written communication skills of the intern.	✓				

7. Please suggest the requisite modification in the curriculum / other feedback to make the students industry ready:

INTRODUCTION OF ORGANIZATION

Larsen & Toubro (L&T) Construction is one of India's largest and most diversified engineering, procurement and construction (EPC) organisations, with operations spanning Buildings & Factories, Transportation Infrastructure, Heavy Civil Infrastructure, Water & Effluent Treatment, and Power Transmission & Distribution. It is widely regarded as a pioneer of large-scale, technology-driven construction in India, executing projects across the country and in several international markets.

The organization places strong emphasis on process standardization and digitization across its project sites, given the scale and geographical spread of its operations. To support this, L&T has developed an extensive internal IT ecosystem that connects its design offices, project sites, procurement teams and finance departments, allowing real-time visibility of project status regardless of location.

A key part of this ecosystem is the Enterprise Information Portal (EIP) — an in-house, web-based end-to-end platform that integrates functions across the Design and Construction phases of a project. EIP consolidates annual budgeting, job performance monitoring, e-procurement, asset management and financial accounting into a single system covering a large number of project sites, enabling the organization to manage its diverse and geographically scattered projects in a coordinated manner.

I undertook my internship with the Water & Effluent Treatment division of L&T Construction, at the Information System Department, Delhi, where I was attached to the team responsible for orienting new users to this platform. Here, I was introduced to the structure, modules and day-to-day usage of EIP as it applies to ongoing construction projects.

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CHAPTER 1: INTRODUCTION

1.1 About the Internship

I underwent a Summer Internship at Larsen & Toubro (L&T) Construction, Water & Effluent Treatment, from 15th June 2026 to 12th August 2026, as part of the Summer Training requirement for the B.Tech. programme in Artificial Intelligence and Data Science at Vivekananda Institute of Professional Studies – Technical Campus, New Delhi. The internship was oriented towards understanding L&T's in-house Enterprise Information Portal (EIP), which is used across the organization's construction projects for managing engineering, procurement, budgeting and project-monitoring workflows.

Rather than being assigned to a single fixed role, I was given exposure across multiple functional areas of EIP, allowing me to build a broad understanding of how a large EPC organization uses enterprise software to run its day-to-day operations across geographically distributed project sites.

1.2 Objective of the Training

The primary objectives of this training were to gain first-hand exposure to a real corporate IT environment, to understand how enterprise-level software systems are designed and used to support large-scale construction operations, and to develop a working understanding of the EIP platform including its modules, workflows and the business logic behind them. The training was also aimed at developing professional skills such as documentation, observation, and effective communication with mentors and team members in a corporate setting.

1.3 Scope of the Report

This report is organized into six chapters. Chapter 2 introduces the Enterprise Information Portal (EIP) and its role within L&T's operations. Chapter 3 describes in detail the key modules and workflows explored during the internship — cost and budget management, access control, supply chain management, the subcontractor module, and the support/ticketing ecosystem. Chapter 4 covers the tools and technologies encountered, including AI-based assistants and the newer EIP 4.0 platform. Chapter 5 discusses key learnings and observations. Chapter 6 presents a project on employee attrition prediction using machine learning, followed by an overall conclusion and list of references.

CHAPTER 2: OVERVIEW OF EIP — ENTERPRISE INFORMATION PORTAL

2.1 What is EIP

EIP (Enterprise Information Portal) is L&T's in-house, web-based end-to-end platform that integrates the Design and Construction phases of a project. It brings together functions such as cost estimation and budgeting, job performance monitoring, supply chain management, subcontractor management, tendering and internal support/ticketing on a single system covering a large number of project sites. Being web-based, EIP can be accessed by authorized users across different project locations, allowing the organization to maintain a single, consistent source of information for each project regardless of where it is being executed.

2.2 Core Functional Areas

During the internship, EIP's functionality was explored under five broad functional areas: (i) Cost and Budget Management, covering cost estimation, monthly job cost reporting and budget tracking; (ii) Supply Chain Management (SCM), covering the procurement lifecycle from purchase request to material receipt; (iii) Subcontractor Management, covering service requests, work orders and subcontractor billing; (iv) Tendering, covering the process of inviting and evaluating bids from vendors and contractors; and (v) Support and Maintenance, covering the ticketing system, the MyLTC App for system status, and an AI-based Virtual Agent for user assistance. Each of these is described in detail in Chapter 3.

2.3 Purpose & Business Value

Since L&T executes a large number of projects simultaneously across different regions, having a single integrated platform like EIP helps standardize processes across sites, gives management real-time visibility into project cost and progress, reduces dependence on manual, paper-based processes, and speeds up approvals for procurement, budgeting and subcontractor payments. This ultimately supports faster decision-making and more transparent tracking of project health at every stage from design through construction.

CHAPTER 3: KEY MODULES AND WORKFLOWS EXPLORED

3.1 Cost & Budget Management

A major part of the internship focused on how EIP tracks project cost from initiation to completion. The key terms and concepts covered were:

- ACE (Accepted Cost Estimation) — a one-time cost estimate prepared and accepted at the start of a project, which acts as the baseline against which future performance is measured.
- JCR (Job Cost Report) — a monthly report that compares actual project cost and progress against the accepted baseline. Each JCR has a validity period of 15 days, within which it must be reviewed and acted upon.
- ETC (Estimate to Complete) — a forward-looking estimate of the cost still required to finish the remaining work on a project.

Project costs are further classified into Direct Cost — covering Plant & Machinery (P&M), Material, and Subcontract (S/C) expenses — and Indirect Cost — covering staff, contingency, welfare, stationery and other miscellaneous overheads.

Other related concepts studied included Resource Planning, Budget Flow (how sanctioned budgets flow down to site-level spending), Committed Cost (cost already committed through purchase orders or work orders even if not yet spent), Cost Overrun (spending beyond the accepted estimate), and how Opportunity and Contingency amounts are set aside while planning a project budget.

3.2 Access Control in EIP

EIP manages user permissions through three access types: Default access, which is granted to all users automatically along with their login; Standard Access, which is based on a user's functional role in the organization; and Delegated Access, which allows a user to temporarily assign their access or responsibilities to another person, for example while on leave.

3.3 Supply Chain Management (SCM)

The SCM module in EIP manages the end-to-end procurement process, divided into two categories: Purchase Orders for material procurement, and Service Orders for services or labour. The standard procurement workflow follows these stages: Purchase Request Creation (PRC) is raised by the site team, followed by a Request for Quote (RFQ) sent to vendors, who respond with a Bidder Quote (BQ). This is followed by preparation of a Techno-Commercial Document (TCD) and a Techno-Commercial Comparison (TCC) of vendor bids, after which a Purchase Order (PO) is released to the selected vendor/supplier or contractor/subcontractor.

Once material is dispatched, an Advance Shipping Note (ASN) is raised by the supplier. On arrival at site, a Goods Inward Note (GIN) records the receipt of goods, followed by an Inspection stage, and finally a Material Receipt Note (MRN) confirms acceptance of the material. Any problems are logged as an Issue in the system for follow-up.

Vendors, suppliers, contractors and subcontractors interact with L&T through a dedicated Partner Portal (PP) — a secure online platform where business partners can view purchase orders and invoices, submit invoices, track payments, and participate in tenders and bids.

A related concept is the Tender — a formal invitation issued by the company to suppliers or contractors to submit bids for supplying goods, services, or executing a project, also referred to as a Tender/RFQ in the system.

3.4 Subcontractor Module

This module manages work assigned to subcontractors, starting with a Service Request (SR) raised for the required work. A Techno-Commercial Document (TCD) is then routed for approval through a defined hierarchy — from the Planning Engineer, to the Project Manager, to the Site Accountant — before being fully approved. A Work Order Document (WOD) is then issued to the subcontractor.

Billing for subcontractor work follows the sequence Planning, then Measurement of completed work, and then Bill generation. The bill passes through the Project Manager for any Bill Deduction and Authorization, after which it moves to the Invoice Register, undergoes Scrutiny, and is finally cleared for Disbursement (Payment).

3.5 Support, Maintenance & the Virtual Agent

EIP includes a Ticketing system, where a Ticket represents a request, complaint, issue or task raised by a user in the system for action. For app-related support, L&T uses the MyLTC App, which shows System Maintenance status — including Current Status (whether any module currently has issues) and Planned Maintenance (advance notice of scheduled maintenance on any module). For quick day-to-day queries, EIP is supported by a Virtual Agent, an AI-based chatbot that assists users in navigating the system and resolving common queries.

CHAPTER 4: TOOLS, TECHNOLOGIES & EIP 4.0

4.1 AI Tools Integrated with EIP

L&T has been integrating AI-based assistants into its digital ecosystem. Tools such as L&T GPT and Microsoft Copilot are used internally for productivity and query assistance, while Julius.ai is used, through the L&T–Mindtree collaboration (L&TMindtree), for AI-driven support. These tools are aimed at helping employees interact with systems like EIP more efficiently through natural-language queries instead of manually navigating every screen.

4.2 EIP 4.0 — Newer Version and Add-ons

During the internship, I also learned about EIP 4.0, an upgraded version of the portal that introduces several new add-on features on top of the core modules described in Chapter 3. These add-ons include an in-built Calculator, a Discussion Forum for users to raise and resolve queries collaboratively, a Notes feature, a Rate Book for reference pricing, a Floater Search and advanced/alert-based search tools, an integrated Helpdesk, and a Work Board for tracking tasks. Together, these additions are aimed at improving day-to-day usability and adoption of the platform among site and office staff.

4.3 How an EIP Feature / Module is Built — Development Lifecycle

Beyond using EIP as an end-user, I was also given a high-level overview of how new features on the platform are developed, which broadly follows these stages: first, a User Requirement Specification (URS) is documented, capturing what the users need; this is followed by Wireframe Design (WFD), which lays out the user interface and how users will interact with the feature; next, a Technical Specification (TS) is prepared, covering requirements, contact/integration points, and technical details such as APIs and databases; the actual Coding and Development stage then builds the feature based on this specification; this is followed by Testing (Quality Testing and User Acceptance Testing/UAT) to validate the feature before it goes live; and finally, the feature is rolled out through Deployment and made available to users, along with change-management activities to support user adoption.

CHAPTER 5: LEARNINGS & OBSERVATIONS

5.1 Key Takeaways

This internship gave me a clear, practical understanding of how a large EPC organization manages cost, procurement and subcontractor work through a single integrated system. I learned how monthly cost reporting (JCR) keeps project budgets under continuous check, how the procurement lifecycle moves systematically from a purchase request to material receipt, and how subcontractor billing follows a strict approval hierarchy before payment is disbursed. I also observed how access control (default, standard and delegated access) is used to balance convenience with accountability in a system used by thousands of employees, and how newer additions like EIP 4.0's add-ons and AI-based tools (Virtual Agent, Copilot, Julius.ai) are being layered on top of the core system to improve usability.

5.2 Challenges Faced

Since I was not assigned a single fixed role, initially it took some time to see how the different modules — cost management, SCM, subcontractor management, tendering and support — connected to one another. This was addressed through guided sessions with my mentor and by taking detailed notes after each session, which helped me build a complete picture of how these modules work together to support a live construction project by the end of the internship.

CHAPTER 6: PROJECT — EMPLOYEE ATTRITION PREDICTION USING MACHINE LEARNING

6.1 Introduction

Employee attrition refers to the loss of employees from an organisation through resignation or other voluntary exit. Unplanned attrition carries direct costs, such as recruiting and training replacements, and indirect costs, such as loss of experience and disruption to ongoing work. Human Resource (HR) departments therefore have a practical interest in knowing which employees are more likely to leave and which workplace factors are associated with leaving.

This chapter describes a data analysis and machine learning project carried out on an HR employee attrition dataset. The project was implemented in a Jupyter Notebook using Python. It moves through data inspection, data quality checks, exploratory data analysis, statistical testing, preprocessing, training and comparison of three classification models, handling of class imbalance, decision-threshold tuning and interpretation of the final model. Because the dataset is imbalanced, much of the work concerns evaluating the models correctly and not only building them.

6.2 Problem Statement

Given structured information about an employee, including demographic details, job role, compensation, tenure, travel requirements, overtime status and self-reported satisfaction ratings, the task is to predict whether the employee has left the organisation (Attrition = Yes) or has stayed (Attrition = No). This is a binary classification problem.

Two features of the data make the problem harder than it first appears. First, only about 16% of employees in the dataset have left, so the two classes are strongly imbalanced. A model that predicts "No" for every employee would already be about 84% accurate while being of no practical use. Second, many of the predictors are related to one another (for example, age, total working years and monthly income), which affects both model behaviour and the interpretation of individual factors.

6.3 Objectives of the Project

- To inspect the dataset and assess its quality in terms of missing values, duplicate records, constant columns and identifier columns.
- To explore how attrition varies across categorical, ordinal and numerical employee attributes using summary statistics and visualisations.
- To test which attributes show a statistically significant association with attrition, with effect sizes and correction for multiple testing.
- To build and compare classification models — Logistic Regression, Random Forest and XGBoost — for predicting attrition.
- To evaluate the models with measures suited to an imbalanced problem, compare them against a majority-class baseline, and use cross-validation to obtain stable estimates.
- To examine the effect of class weighting and of changing the classification threshold on the ability to detect employees who leave.

- To interpret the final model and identify which factors are associated with a higher or lower predicted likelihood of attrition.

6.4 Tools and Technologies Used

The entire project was carried out in a single Jupyter Notebook. The kernel metadata records Python 3.14.5 as the language version. Table 6.1 lists the libraries that are actually imported and used in the notebook, together with their role. Version numbers are stated only where the notebook itself displays them.

Table 6.1: Tools and libraries used in the project

Tool / Library	Role in the project
Python 3.14.5	Programming language for all analysis and modelling
Jupyter Notebook	Interactive environment in which the analysis was written and executed
pandas	Loading the CSV file, inspection, summary statistics, cross-tabulation and grouping
NumPy	Numerical operations, checking processed arrays for missing values, computing odds ratios
Matplotlib	Histograms, box plots, threshold curves and coefficient plots
Seaborn	Correlation heatmap and the combined box/bar plot figure
scikit-learn	Train–test splitting, one-hot encoding, scaling, pipelines, Logistic Regression, Random Forest, stratified cross-validation, evaluation metrics
XGBoost (version 3.4.1)	Gradient-boosted decision tree classifier
SciPy (version 1.18.0)	Chi-square test of independence and Mann–Whitney U test
statsmodels (version 0.15.0)	Variance Inflation Factor and Benjamini–Hochberg correction for multiple testing

6.5 Dataset Description

The data was loaded from a file named HR-Employee-Attrition.csv. It contains **1,470 employee records and 35 columns**. Of these, 26 columns are stored as integers and 9 as text. Each row represents one employee, and the column Attrition (Yes/No) is the target variable. The column structure and class counts correspond to the publicly available IBM HR Analytics Employee Attrition dataset, which is a synthetic dataset created for analytical practice rather than records of real employees of any particular company. The notebook itself does not state the source of the file.

The target variable is imbalanced: 1,233 employees (83.88%) did not leave and 237 employees (16.12%) left, a ratio of about 5.2 to 1. This single fact shapes most of the later decisions on evaluation and modelling.

After inspection, the columns were grouped by their nature so that each group could be analysed and processed appropriately. Table 6.2 shows this grouping as defined in the notebook.

Table 6.2: Grouping of dataset columns

Group	Columns	Treatment
Target	Attrition	Encoded as 0 (No) and 1 (Yes)
Categorical (7)	BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, OverTime	One-hot encoded
Ordinal (9)	Education, EnvironmentSatisfaction, JobInvolvement, JobLevel, JobSatisfaction, PerformanceRating, RelationshipSatisfaction, StockOptionLevel, WorkLifeBalance	Kept as ordered integer codes
Numerical (14)	Age, DailyRate, DistanceFromHome, HourlyRate, MonthlyIncome, MonthlyRate, NumCompaniesWorked, PercentSalaryHike, TotalWorkingYears, TrainingTimesLastYear, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager	Used as continuous/count values
Excluded (4)	EmployeeCount, EmployeeNumber, StandardHours, Over18	Removed before modelling

The ordinal columns hold ratings or levels on a small integer scale (for example, satisfaction from 1 to 4). They were kept separate from the numerical columns because their values are ordered but not necessarily equally spaced, which affects the choice of statistical test in Section 6.9.

6.6 Project Workflow

The work in the notebook proceeds in two phases. The first phase covers data understanding, statistical analysis and an initial round of modelling using an 80/20 train–test split. The second phase repeats the modelling on a fresh three-way split into development, validation and final-test sets with a revised preprocessing design. Table 6.3 summarises the actual sequence of stages.

Table 6.3: Stages of the project workflow

Stage	Activity	Output
1	Load CSV and inspect rows, columns and data types	Dataset of $1,470 \times 35$ confirmed
2	Data quality checks: missing values, duplicates, unique counts, constant and identifier columns	No missing or duplicate rows; 4 columns marked for removal
3	Exploratory analysis of distributions, skewness, outliers and attrition rates by group	Histograms, box plots, cross-tables, correlation heatmap
4	Statistical testing: VIF, Chi-square, Mann–Whitney U, effect sizes, FDR correction	List of attributes significantly associated with attrition
5	Preprocessing: drop excluded columns, encode target, stratified split, one-hot encoding, scaling	Model-ready feature matrices
6	Train Logistic Regression, Random Forest and XGBoost; evaluate on test set and against baseline	Initial comparison of models
7	5-fold stratified cross-validation with pipelines; class-weighted variants	Stable performance estimates
8	Threshold analysis on out-of-fold probabilities	Selected threshold of 0.35 for Logistic Regression
9	Final Logistic Regression evaluated on the test set; coefficients and odds ratios	Final metrics and interpretation
10	Fresh 70/15/15 split with revised preprocessing; cross-validation and threshold analysis repeated	Confirmation of model choice and threshold

6.7 Data Quality Assessment and Preprocessing

6.7.1 Data Quality Checks

Before any analysis, the dataset was checked for common quality problems. The count of missing values was zero for every one of the 35 columns, and the check for fully duplicated rows also returned zero. Since the dataset is complete, no imputation step was needed.

The number of unique values in each column was then examined. Three columns — **EmployeeCount**, **Over18** and **StandardHours** — contain only a single value for all 1,470 employees (for example, **StandardHours** is 80 in every row). A column that never changes cannot separate leavers from stayers, so these were removed. **EmployeeNumber** has a unique value for every employee, confirming it is a record identifier with no predictive meaning, so it was also removed. After removal, the modelling dataset contained 31 columns: 30 predictors and the target.

6.7.2 Target Encoding and Data Splitting

The target values "No" and "Yes" were mapped to 0 and 1 respectively, giving 1,233 negatives and 237 positives. In the first phase, the data was divided into a training set of 1,176 records (80%) and a test set of 294 records (20%) with a fixed random seed of 42. The split was **stratified** on the target, which means each subset keeps the same proportion of leavers as the full dataset. The resulting training set contained 16.16% positives and the test set 15.99%, confirming that stratification worked.

6.7.3 Encoding of Categorical Variables

Machine learning models require numerical input, so the seven categorical columns were converted using **one-hot encoding**. In this method, each category becomes its own binary column that is 1 when the employee belongs to that category and 0 otherwise. For example, **MaritalStatus** becomes three indicator columns for Divorced, Married and Single. This was preferred over assigning arbitrary numbers because categories such as job role have no natural order. Numerical and ordinal columns were passed through unchanged. The encoder was fitted on the training data only and then applied to the test data, producing 51 features in total.

6.7.4 Feature Scaling

The predictors are measured on very different scales: **MonthlyIncome** runs into the thousands while satisfaction ratings range from 1 to 4. Logistic Regression is sensitive to such differences through its optimiser and its regularisation penalty. Therefore **standardisation** (**StandardScaler**) was applied, which rescales each feature by its standard deviation. As with encoding, the scaler was fitted only on the training data. Tree-based models (Random Forest and XGBoost) do not depend on feature scale, and in the initial phase they were trained on the encoded but unscaled data.

6.7.5 Use of Pipelines to Prevent Data Leakage

Later in the project, encoding, scaling and the model were combined into a single scikit-learn **Pipeline**. The benefit is that during cross-validation the preprocessing steps are re-fitted separately inside each fold, using only that fold's training portion. If the encoder or scaler were fitted once on the whole training set before cross-validation, information from the validation folds would leak into the preprocessing and inflate the scores. The notebook explicitly notes that preprocessing is fitted separately inside every fold, and all cross-validation results reported in Section 6.11 come from this leakage-free setup.

6.8 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was used to understand the shape of each variable and to see, before any modelling, which attributes differ between employees who left and those who stayed.

6.8.1 Distribution of Numerical Variables

Histograms were drawn for all 14 numerical variables (Figure 6.1), and the skewness of each was calculated. Skewness measures how asymmetric a distribution is; a positive value means a long tail to the right.

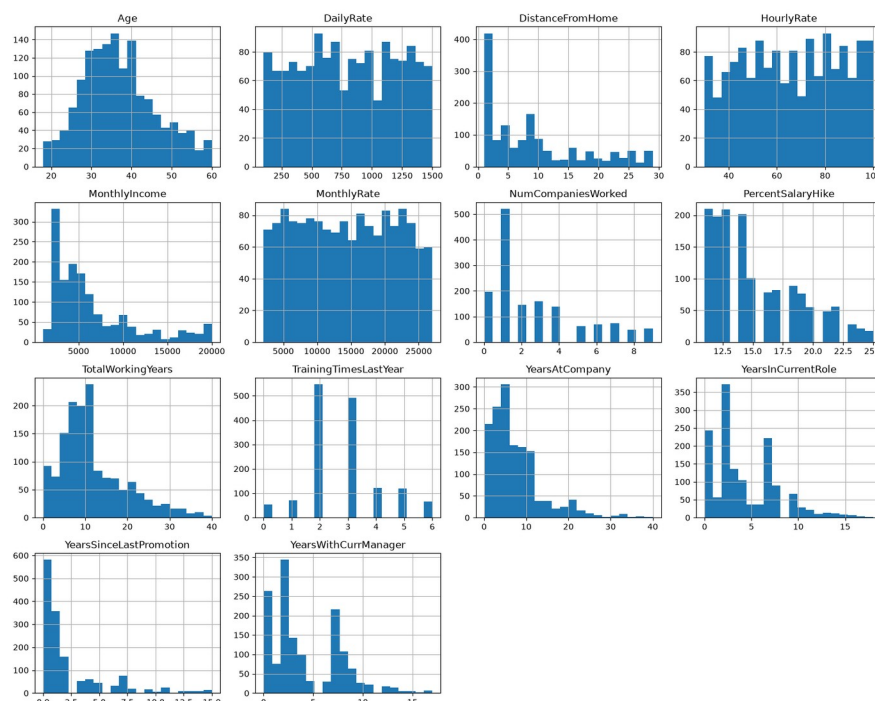


Figure 6.1: Histograms of the fourteen numerical variables

The histograms and skewness values show three distinct patterns. Several tenure and income variables are strongly right-skewed: *YearsSinceLastPromotion* (skewness 1.98), *YearsAtCompany* (1.76), *MonthlyIncome* (1.37), *TotalWorkingYears* (1.12) and *NumCompaniesWorked* (1.03). In each case most employees are concentrated at the lower end, with a small number having very long tenure or high income. Age is only mildly skewed (0.41) and is centred in the 30–40 range. In contrast, *DailyRate*, *HourlyRate* and *MonthlyRate* are almost perfectly flat, with skewness close to zero (−0.004, −0.03 and 0.02). Such flat distributions are unusual for real pay data; later sections show these rate variables are unrelated to attrition and to *MonthlyIncome*.

6.8.2 Outlier Analysis

Box plots of the numerical variables were drawn on a shared axis, but because *MonthlyIncome* and *MonthlyRate* dominate that scale, the smaller variables are compressed near zero and the plot is of limited use. Outliers were therefore counted using the **interquartile range (IQR) rule**, under which a value is flagged if it lies more than 1.5 times the IQR below the first quartile or above the third quartile (Table 6.4).

Table 6.4: Number of IQR-rule outliers per numerical variable

Variable	Outliers	Variable	Outliers
TrainingTimesLastYear	238	YearsInCurrentRole	21
MonthlyIncome	114	YearsWithCurrManager	14
YearsSinceLastPromotion	107	Age, DailyRate, HourlyRate	0
YearsAtCompany	104	MonthlyRate, DistanceFromHome	0
TotalWorkingYears	63	PercentSalaryHike	0
NumCompaniesWorked	52		

The flagged values are plausible rather than erroneous. High incomes belong to senior staff and long tenure to long-serving employees. TrainingTimesLastYear has the highest count only because most employees attended two or three trainings, so 0, 5 and 6 fall outside the narrow IQR. For these reasons **no outliers were removed** in the notebook; they represent genuine variation in the workforce.

6.8.3 Attrition across Categorical Attributes

For each categorical attribute, the attrition rate within each category was computed through cross-tabulation. Table 6.5 lists the highest and lowest rates observed, compared with the overall rate of 16.1%.

Table 6.5: Attrition rate (%) within categories of categorical attributes

Attribute	Categories and attrition rate (%)
OverTime	Yes 30.5; No 10.4
BusinessTravel	Travel_Frequently 24.9; Travel_Rarely 15.0; Non-Travel 8.0
MaritalStatus	Single 25.5; Married 12.5; Divorced 10.1
JobRole	Sales Representative 39.8; Laboratory Technician 23.9; Human Resources 23.1; Sales Executive 17.5; Research Scientist 16.1; Manufacturing Director 6.9; Healthcare Representative 6.9; Manager 4.9; Research Director 2.5
Department	Sales 20.6; Human Resources 19.0; Research & Development 13.8
EducationField	Human Resources 25.9; Technical Degree 24.2; Marketing 22.0; Life Sciences 14.7; Medical 13.6; Other 13.4
Gender	Male 17.0; Female 14.8

Overtime shows the sharpest contrast: employees working overtime left at roughly three times the rate of those who did not. Job role also separates employees strongly, with Sales Representatives leaving at nearly 40% compared with 2.5% for Research Directors. Single employees and frequent travellers also have noticeably higher attrition. The difference between male and female employees is small (about two percentage points), which the statistical test in Section 6.9 later confirms is not significant.

6.8.4 Attrition across Ordinal Attributes

The same analysis was carried out for the ordinal ratings and levels. The clearest trends were:

- **JobInvolvement:** attrition falls steadily from 33.7% at level 1 to 9.0% at level 4.
- **JobLevel:** 26.3% at entry level 1, against 9.7% at level 2 and 4.7% at level 4; level 3 (14.7%) breaks the trend.
- **StockOptionLevel:** 24.4% with no stock options, against 9.4% and 7.6% at levels 1 and 2; level 3 rises to 17.6%.

- **EnvironmentSatisfaction and JobSatisfaction:** the lowest rating (1) has the highest attrition (25.4% and 22.8%), while the other ratings are closer to the overall average.
- **WorkLifeBalance:** a rating of 1 is associated with 31.3% attrition, while ratings 2 to 4 lie between 14% and 18%.
- **PerformanceRating:** almost identical for rating 3 (16.1%) and rating 4, the only two ratings present in the data.

6.8.5 Numerical Variables by Attrition Group

To compare the two groups directly, box plots of each numerical variable were drawn separately for leavers and stayers, and the median and mean of each variable were computed for both groups (Table 6.6). The median was emphasised because it is less affected by the skew noted above. Box plots for the most relevant variables appear in Figure 6.3.

Table 6.6: Median values of selected numerical variables by attrition group

Variable	Stayed (No)	Left (Yes)
Age (years)	36	32
MonthlyIncome	5,204	3,202
TotalWorkingYears	10	7
YearsAtCompany	6	3
YearsInCurrentRole	3	2
YearsWithCurrManager	3	2
DistanceFromHome	7	9
TrainingTimesLastYear	3	2
HourlyRate	66	66
PercentSalaryHike	14	14

Employees who left are, as a group, younger, earn less, have fewer years of total experience and have spent less time at the company, in their current role and with their current manager. The median monthly income of leavers is about 38% lower than that of stayers. Leavers also tend to live slightly further from work. By contrast, HourlyRate and PercentSalaryHike have identical medians in both groups, which suggests they carry little information about attrition.

6.8.6 Correlation Analysis

A Pearson correlation matrix of the numerical variables was computed and displayed as a heatmap (Figure 6.2). Correlation values range from -1 to $+1$ and describe the strength of the linear relationship between two variables.

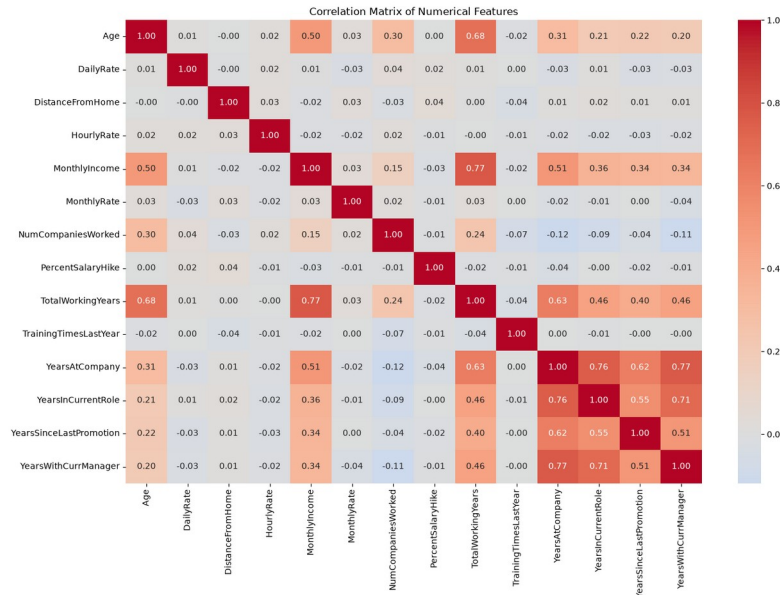


Figure 6.2: Correlation heatmap of numerical variables

The heatmap shows a block of related experience and tenure variables. The strongest relationships are between MonthlyIncome and TotalWorkingYears (0.77), YearsAtCompany and YearsWithCurrManager (0.77), YearsAtCompany and YearsInCurrentRole (0.76), YearsInCurrentRole and YearsWithCurrManager (0.71), and Age and TotalWorkingYears (0.68). DailyRate, HourlyRate and MonthlyRate show almost no correlation with anything, including MonthlyIncome, which reinforces the observation from Section 6.8.1 that these rate columns behave like unrelated values.

6.8.7 Summary Visualisation of Key Attributes

After the statistical tests, a combined figure was prepared for the attributes that showed the strongest relationships with attrition (Figure 6.3).

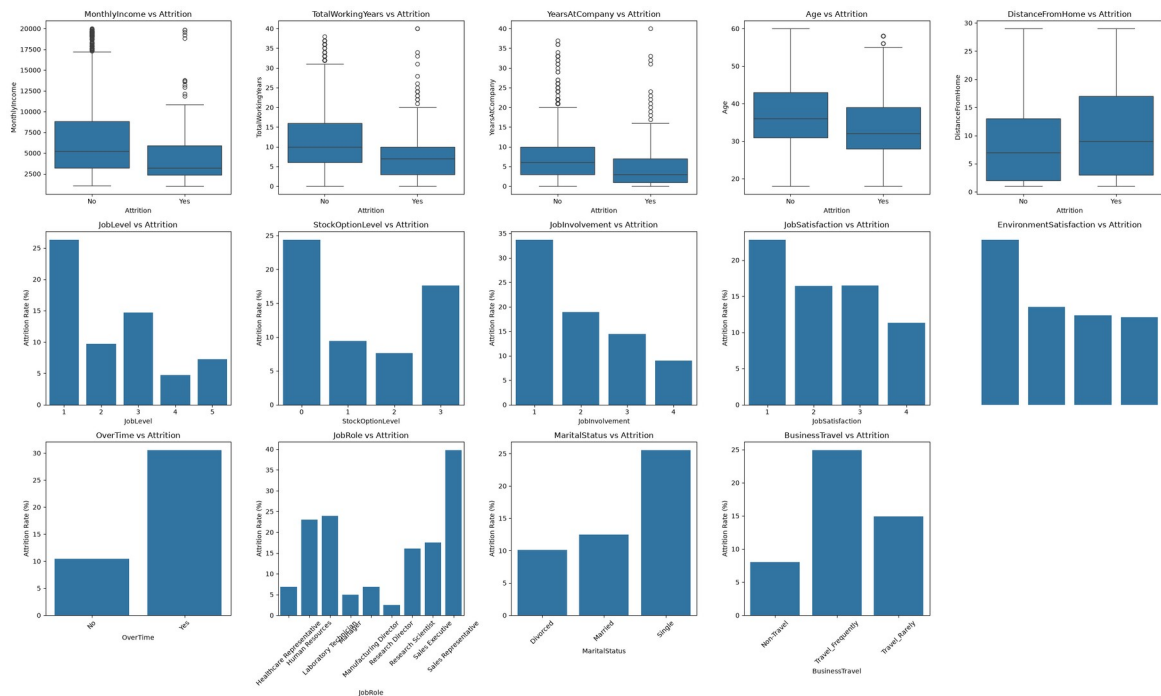


Figure 6.3: Key numerical, ordinal and categorical attributes compared against attrition

The figure brings together the main EDA findings in one view. Two presentation issues should be noted. The EnvironmentSatisfaction panel (middle row, right) was rendered without its axes because of the way unused subplots were hidden, and the JobRole labels in the bottom row overlap. The underlying values are nevertheless those reported in Section 6.8.4.

6.9 Statistical Analysis

Descriptive comparisons show differences between groups but do not indicate whether those differences could have arisen by chance. The notebook therefore applied formal statistical tests, measured effect sizes, and corrected for the fact that many tests were run at once. A significance level of $\alpha = 0.05$ was used throughout.

6.9.1 Multicollinearity Check using Variance Inflation Factor

The **Variance Inflation Factor (VIF)** measures how much the variance of a regression coefficient is inflated because a predictor can be explained by the other predictors. A VIF of 1 indicates no overlap; values above about 5 to 10 are usually treated as a sign of serious multicollinearity. VIF was calculated for the 14 numerical variables.

Table 6.7: VIF of numerical variables

Variable	VIF	Variable	VIF
TotalWorkingYears	4.65	YearsSinceLastPromotion	1.67
YearsAtCompany	4.56	NumCompaniesWorked	1.24
YearsWithCurrManager	2.75	DailyRate	1.01
YearsInCurrentRole	2.68	MonthlyRate	1.01
MonthlyIncome	2.53	DistanceFromHome	1.01
Age	1.99	Other three variables	≈ 1.00

The highest values belong to TotalWorkingYears (4.65) and YearsAtCompany (4.56), reflecting the tenure block seen in the heatmap. Since no variable exceeds 5, the notebook records the decision that no numerical feature would be removed on the basis of VIF. The overlap is moderate and is explained by the natural relationship between different measures of work history.

6.9.2 Chi-square Test and Cramér's V for Categorical Attributes

The **Chi-square test of independence** checks whether two categorical variables are related. The null hypothesis is that the attribute and attrition are independent; a small p-value indicates that the observed pattern of counts is unlikely under independence. Because a p-value does not say how strong the relationship is, **Cramér's V** was also calculated. It ranges from 0 (no association) to 1 (perfect association).

Table 6.8: Chi-square test and Cramér's V for categorical attributes

Attribute	χ^2 statistic	df	p-value	Cramér's V
OverTime	87.56	1	8.2×10^{-21}	0.244
JobRole	86.19	8	2.8×10^{-15}	0.242
MaritalStatus	46.16	2	9.5×10^{-11}	0.177
BusinessTravel	24.18	2	5.6×10^{-6}	0.128
Department	10.80	2	0.0045	0.086
EducationField	16.02	5	0.0068	0.104
Gender	1.12	1	0.291	0.028

Six of the seven categorical attributes are significantly associated with attrition, while Gender is not. OverTime and JobRole have the largest effect sizes (about 0.24), which in practical terms is a modest but meaningful association. Department and EducationField are statistically significant but their effect sizes are small.

6.9.3 Mann–Whitney U Test and Rank-Biserial Correlation

For the ordinal and numerical variables, the **Mann–Whitney U test** was used to compare leavers and stayers. It is a non-parametric test: it compares the ranks of values in the two groups instead of their means, and does not assume normally distributed data. This made it appropriate here because many numerical variables are skewed (Section 6.8.1) and the ordinal ratings are not truly continuous. The effect size was measured by the **rank-biserial correlation**, which ranges from -1 to $+1$. In the way it was calculated in the notebook, a positive value means that employees who stayed tend to have higher values than those who left.

Table 6.9: Mann–Whitney U test results with rank-biserial correlation (r), $p < 0.05$ before correction

Variable	Type	p-value	r
TotalWorkingYears	Numerical	2.4×10^{-14}	0.312
MonthlyIncome	Numerical	3.0×10^{-14}	0.311
YearsAtCompany	Numerical	2.9×10^{-13}	0.298
JobLevel	Ordinal	3.0×10^{-13}	0.283
YearsInCurrentRole	Numerical	4.4×10^{-12}	0.280
YearsWithCurrManager	Numerical	1.8×10^{-11}	0.272
Age	Numerical	5.3×10^{-11}	0.269
StockOptionLevel	Ordinal	4.0×10^{-11}	0.250
JobInvolvement	Ordinal	4.7×10^{-6}	0.165
JobSatisfaction	Ordinal	8.0×10^{-5}	0.156
EnvironmentSatisfaction	Ordinal	2.2×10^{-4}	0.146
DistanceFromHome	Numerical	0.0024	-0.124
DailyRate	Numerical	0.029	0.089
YearsSinceLastPromotion	Numerical	0.041	0.080
WorkLifeBalance	Ordinal	0.046	0.071
TrainingTimesLastYear	Numerical	0.047	0.078

The remaining seven variables were not significant even before correction: RelationshipSatisfaction ($p = 0.10$), NumCompaniesWorked, Education, PercentSalaryHike, MonthlyRate, HourlyRate and PerformanceRating ($p = 0.91$), all with $|r|$ below 0.07.

The strongest effects (r around 0.27 to 0.31) belong to experience, income, tenure and age, followed by JobLevel and StockOptionLevel. All of these are positive, meaning stayers have more experience, higher pay, longer tenure, higher job level and more stock options. DistanceFromHome is the only notable negative effect: leavers tend to live further away. The satisfaction and involvement ratings are significant with smaller effects.

6.9.4 Correction for Multiple Testing

In total, 30 separate tests were carried out (7 Chi-square and 23 Mann–Whitney U tests). When many tests are run at $\alpha = 0.05$, some will appear significant purely by chance. To control this, the

Benjamini–Hochberg procedure, which controls the false discovery rate (FDR), was applied jointly to all 30 p-values.

After correction, **19 attributes remained significant**. In order of adjusted p-value, these are: OverTime, JobRole, MonthlyIncome, TotalWorkingYears, JobLevel, YearsAtCompany, YearsInCurrentRole, YearsWithCurrManager, StockOptionLevel, Age, MaritalStatus, JobInvolvement, BusinessTravel, JobSatisfaction, EnvironmentSatisfaction, DistanceFromHome, Department, EducationField and DailyRate. DailyRate only just passed (adjusted $p = 0.046$). Three attributes that were significant before correction — YearsSinceLastPromotion, WorkLifeBalance and TrainingTimesLastYear — lost significance after correction (adjusted $p \approx 0.06$), so their association with attrition should be treated as weak evidence at best.

These tests describe each attribute on its own. An attribute that is not significant alone may still matter in a model once other attributes are accounted for, a point that returns in Section 6.12.

6.10 Machine Learning Models Used

Three classification algorithms were trained and compared. They were chosen to represent different modelling approaches: a linear model and two tree-based ensemble models.

6.10.1 Logistic Regression

Logistic Regression is a linear model for binary classification. It computes a weighted sum of the input features and passes it through the logistic (sigmoid) function, which converts the result into a probability between 0 and 1. Each weight, or coefficient, indicates how the log-odds of attrition change when that feature increases, with all other features held fixed. A label is assigned by comparing the probability with a threshold, 0.5 by default. It was trained with up to 1,000 iterations and scikit-learn's default L2 regularisation, and was included because its coefficients can be interpreted directly.

6.10.2 Random Forest

Random Forest is an ensemble of decision trees. Each tree is trained on a random bootstrap sample of the training data and considers a random subset of features at each split. The final prediction combines the votes of all trees. Averaging reduces overfitting and lets the model capture non-linear effects and interactions. The first phase used 100 trees and the second phase used 300 trees, both with a fixed random seed.

6.10.3 XGBoost

XGBoost (Extreme Gradient Boosting) also builds an ensemble of decision trees, but sequentially: each new tree is fitted to correct the errors of the trees built before it. The model was configured with 100 trees and log-loss as the evaluation metric. For the class-weighted variant, its `scale_pos_weight` parameter was set to the ratio of negative to positive examples in the training data (about 5.19), which increases the penalty for misclassifying leavers.

No systematic hyperparameter search (such as grid search) was performed for any of the three models. Apart from the settings described above, default parameters were used.

6.11 Model Training and Evaluation

6.11.1 Choice of Evaluation Metrics

Because only about one in six employees left, accuracy alone would be misleading. The notebook therefore reported a set of complementary metrics, where the positive class is "left the organisation":

- **Accuracy:** the share of all predictions that are correct.
- **Precision:** of the employees predicted to leave, the share who actually left.
- **Recall:** of the employees who actually left, the share the model correctly identified.
- **F1-score:** the harmonic mean of precision and recall, which is high only when both are reasonably high.
- **ROC-AUC:** how well the model ranks leavers above stayers across all thresholds (0.5 is random).
- **PR-AUC (Average Precision):** area under the precision–recall curve; more informative than ROC-AUC when positives are rare. A random model scores about 0.16 here.

PR-AUC was used as the main criterion for ranking models, and F1-score was used to select the decision threshold.

6.11.2 Majority-Class Baseline

To give the results a point of reference, a baseline that predicts "No attrition" for every employee was evaluated on the test set. It achieved an **accuracy of 84.0%**, with precision, recall and F1-score all equal to zero, since it never identifies a single leaver. This baseline shows that any accuracy figure around 84% on this dataset is not in itself evidence of a useful model.

6.11.3 Initial Results on the Test Set

In the first round, each model was trained on the 1,176 training records and evaluated on the 294 test records (247 stayers and 47 leavers) at the default threshold of 0.5. Table 6.10 shows the results.

Table 6.10: Initial test-set results at threshold 0.5

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Majority baseline	0.840	0.000	0.000	0.000	—	—
Logistic Regression	0.861	0.615	0.340	0.438	0.808	0.567
Random Forest	0.844	0.533	0.170	0.258	0.799	0.492
XGBoost	0.857	0.619	0.277	0.382	0.736	0.450

All three models barely exceed the baseline in accuracy, but they differ considerably in their ability to find leavers. Logistic Regression identified 16 of the 47 leavers (recall 0.34), XGBoost 13 and Random Forest only 8. In every case most leavers were missed, which is the typical behaviour of models trained on imbalanced data at the default threshold. Logistic Regression achieved the best F1, ROC-AUC and PR-AUC.

6.11.4 Stratified Cross-Validation

A single test set of 294 records, with only 47 positives, can give results that depend heavily on which employees happen to fall into it. To obtain more stable estimates, **5-fold stratified cross-**

validation was performed on the training set. The training data is divided into five folds with equal class proportions; each model is trained on four folds and evaluated on the fifth, and this is repeated five times so that every record is used once for evaluation. As described in Section 6.7.5, preprocessing was fitted inside each fold through a pipeline.

Table 6.11: Mean 5-fold cross-validation results on the training set (Phase 1)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.887	0.746	0.442	0.552	0.837	0.644
XGBoost	0.875	0.754	0.347	0.468	0.794	0.556
Random Forest	0.866	0.865	0.216	0.337	0.804	0.553
Logistic Regression (Balanced)	0.749	0.363	0.721	0.481	0.823	0.600
XGBoost (Balanced)	0.871	0.656	0.416	0.508	0.796	0.574
Random Forest (Balanced)	0.862	0.627	0.374	0.458	0.813	0.561

The cross-validation results confirm the ranking from the single test split. Logistic Regression has the highest PR-AUC (0.644), ROC-AUC (0.837) and F1 (0.552). Random Forest has the highest precision (0.865) but finds only about one in five leavers. The flexibility of the tree ensembles does not appear to help at this sample size.

6.11.5 Handling Class Imbalance through Class Weighting

One way to make a model pay more attention to the minority class is **class weighting**, which increases the cost of misclassifying leavers during training. For Logistic Regression and Random Forest, the "balanced" setting was used, which weights each class inversely to its frequency. For XGBoost, `scale_pos_weight` was set to 5.19. No resampling technique (such as oversampling or SMOTE) was used; imbalance was addressed through weighting and, in the next step, through threshold adjustment.

The lower half of Table 6.11 shows the effect. Balanced Logistic Regression raised recall from 0.442 to 0.721, but precision fell from 0.746 to 0.363, and its PR-AUC (0.600) was lower than that of the unweighted model. For the tree models, weighting improved recall and F1 modestly. The key observation is that class weighting mainly shifts the balance between precision and recall rather than improving the model's underlying ability to rank leavers above stayers. The same shift can be obtained, with more control, by changing the classification threshold.

6.11.6 Decision Threshold Tuning

A classifier outputs a probability, and the label depends on the threshold chosen. Lowering the threshold below 0.5 labels more employees as likely to leave, which increases recall at the cost of precision. To choose a threshold without using the test set, **out-of-fold probabilities** were generated: each training record received a predicted probability from a model that had not seen it during training. Thresholds from 0.20 to 0.70 in steps of 0.05 were then applied to these probabilities, and precision, recall and F1 were calculated at each step for three candidate models.

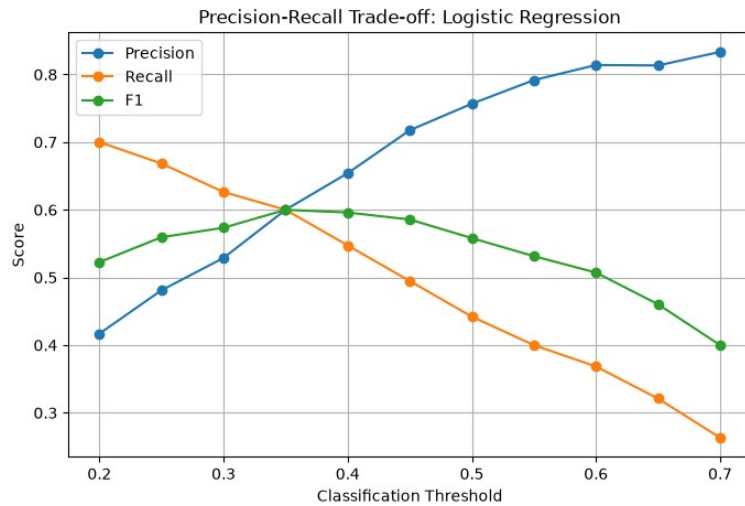


Figure 6.4: Precision, recall and F1-score against classification threshold for Logistic Regression

Figure 6.4 shows the expected trade-off for Logistic Regression: as the threshold rises, precision increases steadily (from 0.42 at 0.20 to 0.83 at 0.70) while recall falls (from 0.70 to 0.26). F1 forms a broad peak and reaches its maximum of **0.600 at a threshold of 0.35**, where precision and recall are both 0.600. At the default 0.5 threshold, the F1-score is lower (0.558) because recall drops to 0.442.

The notebook produced the same plot for the two class-weighted models, where the curves are shifted. Balanced Logistic Regression reaches its best F1 of 0.551 only at the highest threshold tested (0.70), which is consistent with class weighting having already pushed its probabilities upward. Balanced XGBoost reaches its best F1 of 0.526 at the lowest threshold tested (0.20). Since both optimal points lie at the edge of the tested range, the true optimum for these two models may lie outside it. In any case, neither exceeded the unweighted Logistic Regression at its best threshold, as summarised in Table 6.12.

Table 6.12: Best threshold for each candidate model (out-of-fold, training set)

Model	Best threshold	Precision	Recall	F1
Logistic Regression	0.35	0.600	0.600	0.600
Logistic Regression (Balanced)	0.70	0.526	0.579	0.551
XGBoost (Balanced)	0.20	0.498	0.558	0.526

6.11.7 Final Model Evaluation on the Test Set

Based on the cross-validation and threshold analysis, the final model selected in the first phase was **unweighted Logistic Regression with a threshold of 0.35**. It was retrained within a pipeline on the full training set and evaluated on the test set.

Table 6.13: Final Logistic Regression results on the test set (threshold 0.35)

Metric	Value	Metric	Value
Accuracy	0.837	F1-score	0.467
Precision	0.488	ROC-AUC	0.808
Recall	0.447	PR-AUC	0.567

Table 6.14: Confusion matrix of the final model on the test set

	Predicted: Stayed	Predicted: Left	Total
Actual: Stayed	225	22	247
Actual: Left	26	21	47

The final model correctly flagged 21 of the 47 employees who left (recall 0.447), compared with 16 at the default threshold. The cost is an increase in false alarms from 10 to 22, so that about half of the employees it flags actually left (precision 0.488). Overall accuracy fell slightly to 0.837, which is marginally below the majority baseline of 0.840. The baseline reaches its accuracy by never identifying a leaver; the tuned model gives up a few correct "stay" predictions to detect more departures. The ROC-AUC of 0.808 and PR-AUC of 0.567 (against a random-level PR-AUC of about 0.16) show that the model ranks employees by attrition risk far better than chance.

The test-set F1-score (0.467) is lower than the cross-validated F1 at the same threshold (0.600). With only 47 positive cases, a few predictions change the metrics noticeably, but the gap also suggests the cross-validated figures are somewhat optimistic.

6.12 Model Interpretation

An advantage of Logistic Regression is that its coefficients can be examined directly. The coefficients of the final model were extracted and ranked by absolute size. Figure 6.5 shows the fifteen largest.

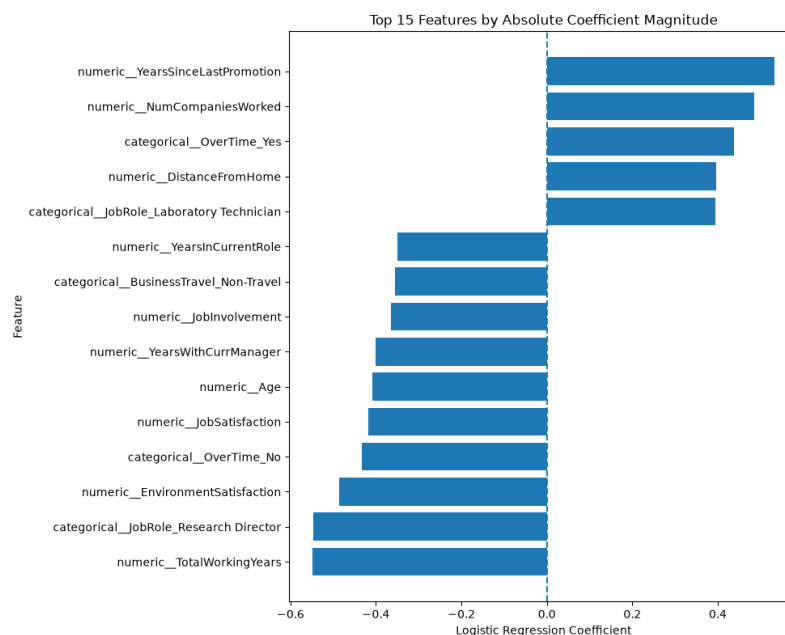


Figure 6.5: Top fifteen Logistic Regression coefficients by absolute magnitude

A positive coefficient means that a higher value of the feature is associated with a higher predicted probability of attrition, and a negative coefficient means the opposite. The coefficients were also converted to **odds ratios** by exponentiation. Because the features were standardised before training, each odds ratio refers to a one-standard-deviation increase in a numerical or ordinal feature (or the equivalent scaled change for an indicator column), with other features held fixed. An odds ratio above 1 indicates higher odds of leaving.

Table 6.15: Features with the largest coefficients in the final model

Higher predicted attrition	Odds ratio	Lower predicted attrition	Odds ratio
YearsSinceLastPromotion	1.70	TotalWorkingYears	0.58
NumCompaniesWorked	1.62	JobRole: Research Director	0.58
OverTime: Yes	1.55	EnvironmentSatisfaction	0.61
DistanceFromHome	1.49	OverTime: No	0.65
JobRole: Laboratory Technician	1.48	JobSatisfaction	0.66
JobRole: Sales Representative	1.39	Age	0.67
BusinessTravel: Travel_Frequently	1.39	YearsWithCurrManager	0.67
JobRole: Human Resources	1.39	JobInvolvement	0.69
JobLevel	1.35	BusinessTravel: Non-Travel	0.70
MaritalStatus: Single	1.27	YearsInCurrentRole	0.70

Many of these results agree with the exploratory and statistical findings. Overtime, frequent travel, longer commuting distance, single marital status and roles such as Laboratory Technician and Sales Representative are associated with higher predicted attrition. Greater total experience, older age, higher environment and job satisfaction, higher job involvement and longer time with the current manager and in the current role are associated with lower predicted attrition.

Three results differ from the single-variable analysis. YearsSinceLastPromotion and NumCompaniesWorked have the two largest positive coefficients although neither was significant on its own after FDR correction; once overall tenure is held constant in the model, a longer wait for promotion or a history of changing employers becomes informative. JobLevel has a positive coefficient even though entry-level staff had the highest attrition, most likely because it overlaps with income and experience, which the model already includes. The coefficients describe associations within this model, not causal effects.

6.13 Refined Workflow with a Three-Way Data Split

In the first phase the test set was used both to compare the initial models and to report the final result. Since a test set should strictly be used once, at the end, the last part of the notebook starts again from a fresh split of the full dataset:

- **Development set:** 1,029 records (70%), containing 863 stayers and 166 leavers, used for cross-validation and threshold selection.
- **Validation set:** 220 records (15%), held out for confirming the chosen configuration.
- **Final test set:** 221 records (15%), reserved for a single final evaluation.

All splits were stratified and used a new random seed (2026). The preprocessing was also revised. One-hot encoding now drops the first category of each categorical variable, which removes the redundancy of having, for example, both OverTime_Yes and OverTime_No columns. Numerical and ordinal columns are standardised with full centring and scaling. This produced 44 features instead of 51. The preprocessor is fitted only on the development set.

Table 6.16: 5-fold cross-validation results on the development set (Phase 2)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.885	0.744	0.446	0.555	0.835	0.652
Random Forest (300 trees)	0.853	0.744	0.168	0.267	0.797	0.541
XGBoost	0.868	0.718	0.320	0.437	0.777	0.533
Logistic Regression (Balanced)	0.773	0.395	0.759	0.519	0.827	0.611
XGBoost (Balanced)	0.863	0.622	0.404	0.485	0.797	0.549
Random Forest (Balanced)	0.852	0.594	0.332	0.418	0.778	0.513

For Logistic Regression, the fold-to-fold standard deviation was 0.009 for accuracy but 0.056 for precision and 0.051 for recall, again reflecting the small number of leavers per fold.

The second phase reproduces the findings of the first on different data partitions. Unweighted Logistic Regression again has the highest PR-AUC, ROC-AUC and F1. Threshold analysis on out-of-fold probabilities again selected a **threshold of 0.35**, where precision, recall and F1 all equal 0.596. The best threshold for balanced Logistic Regression was again 0.70 (F1 = 0.570). The agreement between the two phases increases confidence that the choice of model and threshold is not an artefact of one particular split.

The notebook ends after the threshold analysis. The chosen model has **not yet been evaluated on the validation set or the final test set** of the three-way split, so no final held-out performance figures from the second phase can be reported. The most recent held-out results available are therefore those in Tables 6.13 and 6.14.

6.14 Discussion of Results

Across all comparisons, the simplest model performed best. Logistic Regression outperformed Random Forest and XGBoost on PR-AUC, ROC-AUC and F1 under a single split, under cross-validation and after a fresh re-split. Class weighting did not improve ranking quality; it only moved the precision–recall balance, which threshold tuning did more directly. Lowering the threshold from 0.5 to 0.35 was the most effective single change for detecting leavers.

In practical terms, the model would let an HR team rank employees by estimated risk. At the chosen threshold it found just under half of the leavers in the test set, and about half of the employees it flagged actually left. That is adequate for prioritising retention conversations, where a false alarm costs little, but not for making decisions about individuals automatically.

6.14.1 Confusion-Matrix Analysis at the Two Thresholds

The confusion matrix of the final model at the chosen threshold of 0.35 is given in Table 6.14. The corresponding matrix at the default threshold of 0.5 is not tabulated in the notebook, but it follows directly from figures that are reported: at that threshold the model identified 16 of the 47 leavers and raised 10 false alarms (Section 6.11.7), which implies 31 missed leavers and 237 correctly identified stayers among the 247. Table 6.17 places the two thresholds side by side. The counts for the 0.5 threshold and all rates in the table are derived from the reported figures; they were not printed in this form in the notebook.

Table 6.17: Final Logistic Regression model at thresholds 0.5 and 0.35 (test set of 294 employees)

Quantity	Threshold 0.5	Threshold 0.35
True positives (leavers correctly flagged)	16	21
False positives (stayers wrongly flagged)	10	22
False negatives (leavers missed)	31	26
True negatives (stayers correctly identified)	237	225
Employees flagged (out of 294)	26	43
Recall (leavers found out of 47)	0.340	0.447
Precision (flagged employees who left)	0.615	0.488
Specificity (stayers correctly identified out of 247)	0.960	0.911
Balanced accuracy (mean of recall and specificity)	0.650	0.679
Accuracy	0.861	0.837
Leaver rate among flagged employees relative to the overall leaver rate of 0.160	3.85 times	3.05 times

Lowering the threshold found 5 more leavers (16 to 21) but produced 12 more false alarms (10 to 22), so each additional leaver found cost about 2.4 additional false alarms. Specificity fell from 0.960 to 0.911, which means the model still recognised about 91% of the employees who stayed. Balanced accuracy rose from 0.650 to 0.679. The majority baseline, with a recall of 0 and a specificity of 1, scores exactly 0.500 on this measure, which shows more clearly than raw accuracy (0.840 for the baseline) that the tuned model carries real information about who leaves.

At the threshold of 0.35 the model flagged 43 of the 294 employees (14.6%), a share close to the 16.0% of leavers in the test set, and 48.8% of the flagged employees had actually left, about three times the overall leaver rate. The flagged group is therefore much richer in leavers than the workforce as a whole, but 26 of the 47 leavers (55.3%) were still not flagged. Whether the additional false alarms are an acceptable price for the additional leavers found depends on the relative cost of a missed departure and of an unnecessary retention conversation, a point taken up again in Section 6.16.

6.14.2 Cross-Validated versus Held-Out Performance

Table 6.18 sets the cross-validated estimates of Logistic Regression from the first phase (Table 6.11) beside its test-set results at both thresholds (Tables 6.10 and 6.13).

Table 6.18: Logistic Regression — cross-validated and test-set performance

Metric	5-fold CV, threshold 0.5	Test set, threshold 0.5	Test set, threshold 0.35
Accuracy	0.887	0.861	0.837
Precision	0.746	0.615	0.488
Recall	0.442	0.340	0.447
F1-score	0.552	0.438	0.467
ROC-AUC	0.837	0.808	0.808
PR-AUC	0.644	0.567	0.567

The two threshold-independent measures are lower on the test set: ROC-AUC is 0.808 against 0.837, and PR-AUC is 0.567 against 0.644. At the same threshold of 0.5, precision (0.615 against 0.746), recall (0.340 against 0.442) and F1-score (0.438 against 0.552) are also lower on the test set. Lowering the threshold to 0.35 brings the test-set recall (0.447) back to the cross-validated level (0.442), but at a lower precision (0.488).

A gap of this kind is not surprising, for two reasons. First, the test set contains only 47 leavers, and the second-phase cross-validation showed fold-to-fold standard deviations of 0.056 for precision and 0.051 for recall for this model (Section 6.13); a difference of about 0.10 in recall between a fold-averaged estimate and a single small test set is of the same order as twice that spread. Second, the threshold of 0.35 was chosen because it gave the highest F1-score on the out-of-fold probabilities, so the out-of-fold F1-score of 0.600 at that threshold is the best value on the search grid and is expected to be somewhat higher than the value on independent data (0.467 on the test set). The test-set figures are therefore the more realistic guide to performance, while the cross-validated figures are best read as evidence for the ranking of the models rather than as forecasts of absolute performance.

6.14.3 Consistency of Results across the Two Phases

The project was run twice on different partitions of the same 1,470 records. Table 6.19 lists the partitions used in each phase.

Table 6.19: Data partitions used in the two phases

Phase	Partition	Records	Share of data	Leavers
1	Training	1,176	80%	190 (16.16%)
1	Test	294	20%	47 (15.99%)
2	Development	1,029	70%	166 (16.13%)
2	Validation	220	15%	Not reported separately
2	Final test	221	15%	Not reported separately

The number of leavers in the first-phase training set (190) and the leaver share of the development set (16.13%) are derived from the totals reported in Sections 6.5, 6.7.2 and 6.13. The validation and final-test sets together contain the remaining 71 leavers (237 – 166); because both were stratified, each is expected to hold roughly 35 of them.

Table 6.20 compares the cross-validated PR-AUC and F1-score of all six model variants in the two phases. Random Forest used 100 trees in the first phase and 300 trees in the second.

Table 6.20: Cross-validated PR-AUC and F1-score of each model in the two phases

Model	PR-AUC, Phase 1	PR-AUC, Phase 2	F1, Phase 1	F1, Phase 2
Logistic Regression	0.644	0.652	0.552	0.555
Random Forest	0.553	0.541	0.337	0.267
XGBoost	0.556	0.533	0.468	0.437
Logistic Regression (Balanced)	0.600	0.611	0.481	0.519
XGBoost (Balanced)	0.574	0.549	0.508	0.485
Random Forest (Balanced)	0.561	0.513	0.458	0.418

Unweighted Logistic Regression is the best variant on both measures in both phases, and it changed by less than 0.01 in PR-AUC (0.644 to 0.652) and by only 0.003 in F1-score (0.552 to 0.555). Balanced Logistic Regression is second on PR-AUC in both phases. The tree-based variants moved more: their PR-AUC fell by between 0.012 and 0.048, and the F1-score of Random Forest fell from 0.337 to 0.267. Because the two phases differ in several respects at once — the partition and its size, the random seed, the encoding (51 against 44 features) and the number of trees — the changes in the tree-based models cannot be assigned to any single cause. The stability

of the top of the ranking across all of these differences is nevertheless the main reason for confidence in the choice of Logistic Regression.

6.14.4 Agreement of Evidence across the Analysis Stages

Sections 6.8 to 6.12 examine the same factors in three different ways: through group rates and medians, through hypothesis tests with effect sizes and correction for multiple testing, and through the coefficients of a multivariable model. Table 6.21 brings together the evidence for the main factors. For numerical and ordinal variables the sign convention of Table 6.9 applies, so a positive rank-biserial correlation r means that stayers have higher values.

Table 6.21: Evidence for the main factors at each stage of the analysis

Factor	Exploratory analysis	Statistical test	Logistic Regression odds ratio
OverTime	Attrition 30.5% (Yes) against 10.4% (No)	Significant after correction; Cramér's V 0.244	1.55 (Yes); 0.65 (No)
JobRole	39.8% for Sales Representatives to 2.5% for Research Directors	Significant after correction; Cramér's V 0.242	1.48 (Laboratory Technician); 1.39 (Sales Representative); 0.58 (Research Director)
BusinessTravel	24.9% (frequent travel) against 8.0% (non-travel)	Significant after correction; Cramér's V 0.128	1.39 (frequent travel); 0.70 (non-travel)
MaritalStatus	25.5% (single) against 10.1% (divorced)	Significant after correction; Cramér's V 0.177	1.27 (single)
TotalWorkingYears	Median 7 (left) against 10 (stayed)	Significant after correction; $r = 0.312$	0.58
Age	Median 32 (left) against 36 (stayed)	Significant after correction; $r = 0.269$	0.67
JobSatisfaction	22.8% attrition at the lowest rating	Significant after correction; $r = 0.156$	0.66
EnvironmentSatisfaction	25.4% attrition at the lowest rating	Significant after correction; $r = 0.146$	0.61
DistanceFromHome	Median 9 (left) against 7 (stayed)	Significant after correction; $r = -0.124$	1.49
MonthlyIncome	Median 3,202 (left) against 5,204 (stayed)	Significant after correction; $r = 0.311$	Not among the largest coefficients
YearsSinceLastPromotion	Strongly right-skewed (skewness 1.98)	Not significant after correction (adjusted $p \approx 0.06$)	1.70
NumCompaniesWorked	Right-skewed (skewness 1.03)	Not significant even before correction	1.62

For most factors the three views agree. Overtime, job role, business travel and marital status show large differences in attrition rates, remain significant after correction, and have odds ratios above 1 for the higher-attrition categories. Total working years, age, job satisfaction and environment satisfaction are lower among leavers, remain significant after correction, and have odds ratios below 1. Two factors stand out because the evidence differs between stages. YearsSinceLastPromotion and NumCompaniesWorked have the largest positive odds ratios (1.70 and 1.62) although neither was significant on its own after correction, and MonthlyIncome, which

had one of the strongest single-variable effects, is not among the largest coefficients, possibly because it overlaps with total working years (correlation 0.77 in Section 6.8.6). Gender showed a small difference in attrition rates (17.0% for male and 14.8% for female employees) and was not significant ($p = 0.291$).

6.14.5 Practical Applications and Use Cases

The results support a small number of realistic uses for a model of this kind, provided its limits are respected:

- **Prioritising retention conversations.** Ranking employees by predicted probability lets an HR team decide whom to speak to first. At the threshold of 0.35, the 43 employees flagged in the test set contained 21 of the 47 leavers, so the flagged group was about three times richer in leavers than the workforce as a whole.
- **Matching the threshold to available capacity.** The threshold curves in Figure 6.4 show the trade-off directly: on the out-of-fold probabilities, a threshold of 0.20 gave a recall of about 0.70 with a precision of about 0.42, while a threshold of 0.70 gave a precision of about 0.83 with a recall of about 0.26. A team with room for only a few conversations would choose a high threshold; a team that can afford many would choose a lower one.
- **Reviewing workplace factors at group level.** Overtime, frequent travel, long gaps since the last promotion, low job satisfaction and low job involvement were all associated with higher predicted attrition. These factors can guide where an organisation looks first when reviewing working conditions, bearing in mind that the associations are not causal effects.

The limits are those set out in Section 6.15: the model finds fewer than half of the leavers, the dataset is most likely synthetic, and any use on real employees would require validation on the organisation's own records, together with attention to privacy and to fairness across employee groups.

6.15 Challenges and Limitations

- **Class imbalance and small sample.** With only 237 leavers overall and 47 in the first test set, accuracy was misleading and a change of a few predictions shifts recall and precision noticeably, so all reported values carry considerable uncertainty.
- **Incomplete final evaluation.** The three-way split was created and the model and threshold were re-selected on the development set, but the evaluation on the validation and final-test sets was not carried out in the notebook.
- **Repeated use of the first test set.** It served both the initial model comparison and the final evaluation, which makes that result slightly optimistic.
- **No hyperparameter tuning.** All models used default or near-default settings. The tree-based models, in particular, might perform differently with tuned depth, learning rate or number of trees.
- **Threshold search range.** For the class-weighted Logistic Regression and XGBoost models, the best F1 occurred at the edges of the tested range (0.70 and 0.20), so their best achievable F1 may not have been found.
- **Interpretation of coefficients.** Coefficients are on standardised scales, are affected by correlated predictors and do not show causation. The first-phase encoding kept every category indicator, which blurs the reading of individual dummy coefficients.

- **Nature of the data.** The dataset appears to be the synthetic IBM HR Analytics dataset and not real organisational data. Patterns found here may not transfer to any specific organisation without validation on its own records.

6.15.1 Challenges Encountered and Measures Used to Address Them

The limitations listed above are those that remain after the work was completed. Several other difficulties arose during the project and were handled within the workflow. Table 6.22 lists each of them with the measure used and the section in which it is described.

Table 6.22: Challenges in the project and the measures used to address them

Challenge	Measure used in the project	Section
Imbalanced classes (16.12% leavers)	Stratified splits, a majority-class baseline, PR-AUC and F1-score as main criteria, class weighting and threshold tuning	6.7.2, 6.11
Data leakage during preprocessing and cross-validation	Encoding and scaling placed inside pipelines and fitted separately within every fold	6.7.5
Unstable estimates from a small test set (47 leavers)	5-fold stratified cross-validation and a second phase on a fresh split	6.11.4, 6.13
Skewed numerical variables and ordinal ratings	Non-parametric Mann–Whitney U test with rank-biserial correlation	6.8.1, 6.9.3
Correlated predictors	VIF check (highest value 4.65); no feature removed; coefficients interpreted with caution	6.9.1, 6.12
Many simultaneous significance tests (30 tests)	Benjamini–Hochberg correction of the false discovery rate	6.9.4
Columns with no predictive value	Removal of three constant columns and one identifier column	6.7.1
Values flagged as outliers by the IQR rule	Retained, because they represent genuine variation in the workforce	6.8.2
Reuse of the first test set	Fresh 70/15/15 split with a final test set reserved for a single evaluation	6.13

Most of the steps between loading the data and reporting the results exist to prevent an over-optimistic reading of the model's performance on a small, imbalanced dataset, which is consistent with the main learning stated in Section 6.17.

6.16 Future Scope

- Complete the second-phase workflow by evaluating the selected model once on the validation set and then once on the final test set, and report these as the definitive results.
- Carry out systematic hyperparameter tuning within cross-validation.
- Extend the threshold search for class-weighted models beyond the 0.20–0.70 range, and consider choosing the threshold according to the relative costs of missed leavers and false alarms rather than F1 alone.
- Apply model-agnostic explanation methods, such as permutation importance or SHAP values, to compare the importance of features across the linear and tree-based models.
- Validate the approach on real, anonymised HR data from an organisation, with appropriate attention to data privacy and to fairness across employee groups, before any practical use.

6.16.1 Proposed Steps for Completing the Refined Evaluation

The first item of the future scope can be carried out in a short sequence of steps that follows from the design of the three-way split described in Section 6.13:

- **Fix the configuration.** Use unweighted Logistic Regression in a leakage-free pipeline, fitted on the development set only, with a threshold of 0.35, and make no further changes after this point.
- **Evaluate once on the validation set of 220 records.** Report precision, recall, F1-score, ROC-AUC and PR-AUC together with the confusion matrix, and compare them with the majority-class baseline computed on the same records. Keep the configuration only if the results are broadly consistent with the cross-validated estimates.
- **Evaluate once on the final test set of 221 records.** Report the same measures as the definitive held-out result and set them beside Tables 6.13 and 6.14 so that the two phases can be compared.
- **Report the uncertainty.** The validation and final-test sets together contain 71 leavers, so each holds only about 35, and the metrics from either set will change noticeably with a few predictions. Confidence intervals or repeated splits would give a fairer picture than single point values.

6.17 Conclusion

This project addressed the problem of predicting employee attrition from structured HR data containing 1,470 employee records and 35 attributes, of whom 16.1% had left. The work was carried out in Python in a Jupyter Notebook.

The data had no missing or duplicate records; three constant columns and one identifier were removed. Employees who left were, as a group, younger, lower paid, less experienced and shorter in tenure, and more likely to work overtime, travel frequently, be single or hold roles such as Sales Representative. Chi-square and Mann–Whitney U tests, with effect sizes and Benjamini–Hochberg correction across 30 tests, confirmed 19 attributes as significantly associated with attrition, while gender, education, performance rating, salary hike and the three pay-rate columns were not.

Three classifiers were compared using stratified cross-validation within leakage-free pipelines. Logistic Regression consistently outperformed Random Forest and XGBoost, with a cross-validated PR-AUC of about 0.64–0.65 and ROC-AUC of about 0.84. Class weighting shifted the balance between precision and recall without improving overall ranking quality, and threshold tuning on out-of-fold predictions selected a threshold of 0.35 in two independent splits. On the held-out test set, the final model reached a recall of 0.447, a precision of 0.488, an F1-score of 0.467, a ROC-AUC of 0.808 and a PR-AUC of 0.567, identifying 21 of 47 leavers compared with none for the majority baseline.

The main learning is that, for an imbalanced problem, the evaluation procedure matters as much as the algorithm: a baseline, positive-class metrics, leakage-free cross-validation and a chosen threshold changed the picture more than a more complex model did. The results are moderate and apply to a dataset that is most likely synthetic; the final held-out evaluation of the refined workflow remains to be completed. Within those limits, the project covers a complete machine learning workflow, from raw data to an interpretable model that ranks employees by attrition risk.

CONCLUSION

This Summer Internship at L&T Construction provided valuable, hands-on exposure to how a large-scale engineering and construction organization manages its operations through an integrated enterprise system — the Enterprise Information Portal (EIP). Through guided learning, I gained an understanding of cost and budget management (ACE, JCR, ETC, direct/indirect costs), supply chain management (the purchase-to-payment cycle, tendering, and the Partner Portal), subcontractor management (service requests, work orders and the billing/disbursement workflow), and the support ecosystem around EIP, including its ticketing system, the MyLTC App, and an AI-based Virtual Agent. I was also introduced to EIP 4.0's newer features and to the broader development lifecycle used to build and roll out such enterprise features.

Even though I was not assigned a single, narrowly defined role, the breadth of exposure across these modules gave me a well-rounded, practical understanding of how enterprise software supports large, complex construction projects. Overall, this training strengthened my appreciation for how information technology underpins operations at scale, and it has been a valuable addition to my academic learning as a B.Tech. Artificial Intelligence and Data Science student.

This report also includes, as Chapter 6, a data analysis and machine learning project on employee attrition prediction, carried out in Python in a Jupyter Notebook on an HR dataset of 1,470 employee records. In that project, Logistic Regression consistently outperformed Random Forest and XGBoost, and the final model at a threshold of 0.35 identified 21 of the 47 employees who left in the held-out test set (recall 0.447, precision 0.488, ROC-AUC 0.808). The project thus complements the enterprise-systems exposure described above with a complete, end-to-end machine learning workflow from raw data to an interpretable model.

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